A numerical reference for the spatial Dysthe equation

Stage one: local reference validation

16 October 2026

Abstract

A bounded local campaign tests the existing water-wave reference before initial-data simulation or operator learning. Analytic controls, independent residuals, invariant audits, and separate step, grid, and domain comparisons qualify the stated periodic problem. Matched propagated controls separate phase and translation drift from deformation. Finite-window persistence establishes neither soliton discovery nor spectral or nonlinear stability.

1 Equation and discretization

The verified project is

`/home/matasvai/Documents/Projects/Dysthe`.

The contract `water-wave-dysthe-spatial-v1` specifies the complex scaled free-surface envelope $u(\xi,\tau)$, propagation distance ξ, and retarded time τ:

$$u_\xi = -\mathrm{i}u_{\tau\tau} - \mathrm{i}|u|^2u - \epsilon\left(8|u|^2u_\tau + 2u^2\overline{u_\tau} + 2\mathrm{i}u|D_\tau|(|u|^2)\right). \tag{1.1}$$

This is equation (1.3) of Fedele and Dutykh [1], with $(a,h,c,e,f) = (1,1,8,2,2)$. The source normalization is $B = \epsilon u$, $\tau = \epsilon(2x-t)$, $\xi = \epsilon^2 x$, and $\epsilon = k_0 a$ in dimensionless carrier coordinates. The $\epsilon = 0$ NLS control is a mathematical limit; the coordinate transformation degenerates there. Complex128 arrays have axes $[\xi,\tau]$, not two horizontal directions.

The domain $I_L = [-L/2, L/2)$ is periodic, with $\tau_j = -L/2 + jL/N$, $0 \le j < N$. No absorbing layer is used. For Fourier kernel $e^{-\mathrm{i}k\tau}$ and $k_m = 2\pi m/L$, ∂_τ has symbol $\mathrm{i}k_m$ and $|D_\tau|$ has symbol $|k_m|$, including a zero mean mode. Thus $|D_\tau| = \mathcal{H}\partial_\tau$ with Hilbert symbol $-\mathrm{i}\,\mathrm{sign}(k_m)$.

Let P_N retain the strict band $|m| < N/4$, let $\mathcal{F}$ be the full RHS of (1.1), and let $\mathcal{L} = -\mathrm{i}\partial_{\tau\tau}$. The computed Galerkin system is

$$u_\xi = \mathcal{L}u + P_N(\mathcal{F}(u) - \mathcal{L}u), \qquad u = P_N u. \tag{1.2}$$

This band resolves retained cubic products without aliasing. Lawson integrating-factor RK4 applies $\exp(\mathrm{i}k_m^2\Delta\xi)$ exactly and advances the transformed nonlinear equation with classical RK4. Petviashvili-type profile construction restricts conjugate reflection; propagation imposes no such symmetry. Iteration convergence does not establish stability or enumerate all branches.

2 Controls and diagnostics

Use $\|f\|_{2,L}^2 = \int_{I_L} |f|^2 \,\mathrm{d}\tau$, discretized by $(L/N)\sum_j |f_j|^2$. Linear multiplier tests and explicit Fourier convolution of a nonconstant complex multimode field check signs, conjugation, and nonzero mean

flow. Exact periodic plane waves satisfy

$$u = Ae^{\mathrm{i}k\tau - \mathrm{i}\Omega\xi}, \qquad \Omega = A^2 - k^2 + 6\epsilon k A^2, \quad kL/(2\pi) \in \mathbb{Z}. \tag{2.1}$$

The implementation's traveling convention is tested at $\epsilon = 0$ by

$$\begin{aligned} u(\xi,\tau) &= e^{-\mathrm{i}\mu\xi} F(\tau - s\xi), \\ F(\tau) &= \sqrt{2\nu}\,\mathrm{sech}(\sqrt{\nu}\,\tau) e^{-\mathrm{i}s\tau/2}, \qquad \nu = \mu - s^2/4 > 0. \end{aligned} \tag{2.2}$$

This whole-line solution requires negligible periodic-window error. Nonzero s tests translation and phase; a moving Dysthe profile supplies an additional control.

For a profile F, let $T = -\mathrm{i}\mu F - sF_\tau \neq 0$. Fourfold Fourier interpolation resolves its full cubic RHS and gives independent residuals

$$\begin{aligned} r_{\mathrm{full}} &= \frac{\|\mathcal{F}(F) - T\|_{2,L}}{\|T\|_{2,L}}, \qquad r_{\mathrm{ret}} = \frac{\|P_N[\mathcal{F}(F) - T]\|_{2,L}}{\|T\|_{2,L}}, \\ r_{\mathrm{disc}} &= \frac{\|(1 - P_N)\mathcal{F}(F)\|_{2,L}}{\|T\|_{2,L}}. \end{aligned} \tag{2.3}$$

These qualify the trigonometric interpolant, not whole-line existence or accumulated propagation error. Arbitrary-field discarded-RHS diagnostics instead normalize by $\|\mathcal{F}(u)\|_{2,L}$ and record that choice.

With $\rho = |u|^2$ and $J = \mathrm{Im}(u_\tau \overline{u})$, monitor

$$\mathcal{A} = \int_{I_L} \rho \, \mathrm{d}\tau, \tag{2.4}$$

$$\mathcal{M} = \int_{I_L} (2J - 2\epsilon\rho^2) \, \mathrm{d}\tau, \tag{2.5}$$

$$\mathcal{E} = \int_{I_L} \left(|u_\tau|^2 - \tfrac{1}{2}\rho^2 + \tfrac{10}{3}\epsilon^2\rho^3 - 5\epsilon\rho J - \epsilon\rho |D_\tau| \rho \right) \mathrm{d}\tau. \tag{2.6}$$

These are continuum functionals [1]; u is noncanonical, and no periodic canonical gauge is imposed. Quadrature on $2N$ interpolated points resolves the degree-six density: $6K < 2N$ for retained cutoff $K < N/4$. An independent directional-derivative audit uses both full and Galerkin RHSs. Projection can change energy/momentum conservation; RK4 can change action. Absolute drift is retained, with relative drift omitted near zero. Conservation alone does not establish accuracy.

For perturbed field w and its matched propagated control v, fit only translation a and constant phase θ:

$$D_{\mathrm{shape}} = \min_{a,\theta} \frac{\left\| w - e^{\mathrm{i}\theta} v(\cdot - a) \right\|_{2,L}}{\|v\|_{2,L}}, \qquad q = e^{-\mathrm{i}\theta} w(\cdot + a) - v. \tag{2.7}$$

All discrete translations are searched, followed by correlation refinement and polishing; continuous global optimality is not proved. Regression controls test roundoff symmetry recovery and preservation of amplitude changes. Raw error, periodic translation, and principal-valued phase remain separate. Comparing signed fields q between refinements retains radiation and deformation that differences of scalar norms would miss.

Step comparisons fix N, L; grid comparisons fix $L, \Delta\xi$; domain comparisons preserve $L/N, \Delta\xi$ and recompute profiles. Common-window errors and larger-domain exterior norms are reported separately. Periodicity affects mean flow even with negligible boundary amplitude. Far-field group speed $-2k$ and folding exterior radiation into a smaller period diagnose wrapping; neither provides a nonlinear error bound. Boundary and spectral-edge action are supporting diagnostics. Weak-amplitude and narrowband screens are design choices, not universal physical bounds.

3 Measured results and decision

3.1 Analytic controls and original-profile audit

The existing test suite passes 40 tests and 34 subtests (`../results/tests.log`), including the explicit multimode convolution check with nonzero mean flow. For plane waves $A = 0.6$, $k = 0, \pm 3$, the largest RHS error is 1.43×10^{-12} and the largest propagated complex error at $\xi = 2$ is 1.45×10^{-13}. The moving NLS control at $N = 2048$, $L = 80$, $\mu = 2$, $s = 1$ has the errors in Table 1. Successive observed orders approach four (Figure 1a). Here $p = \log(e_{\text{coarse}}/e_{\text{fine}})/\log(h_{\text{coarse}}/h_{\text{fine}})$; these are exact-solution errors, not estimates inferred from conservation.

Table 1: Moving NLS complex relative L2 error at $\xi = 2$. The ratio compares with the preceding coarser step.

$\Delta\xi$	Error	Ratio	Order p
0.04	3.79×10^{-4}	—	—
0.02	2.56×10^{-5}	14.767	3.884
0.01	1.65×10^{-6}	15.548	3.959
0.005	1.04×10^{-7}	15.823	3.984

A recomputed moving Dysthe profile ($\epsilon = 0.025$, $\mu = 2$, $s = 1$, $N = 4096$, $L = 160$) has full residual 2.06×10^{-11}. At $\xi = 2$ and $\Delta\xi = 0.000625$, its raw complex error against the declared traveling solution is 9.66×10^{-12}.

On two independent multimode test fields, full-RHS directional rates of $\mathcal{A}, \mathcal{M}, \mathcal{E}$ are at most 3.24×10^{-12} in absolute value per unit ξ, while the near-cutoff Galerkin energy rate is 5.92×10^{-5}. This explicitly exposes a projection contribution to invariant drift.

All 28 original stored profiles were independently re-audited on their original $N = 2048$, $L = 160$ grid (Figure 1b). Although every retained residual is at most 4.36×10^{-11}, 22 full periodic residuals exceed the current 10^{-8} gate. The largest is 6.12×10^{-3} at $(\epsilon, \mu, s) = (0.15, 3, 0)$; 16 profiles also exceed the $\epsilon \max |F| \leq 0.1$ design screen. These results require requalification or finer profile grids; they do not establish physical instability or nonexistence. The CSV audit retains the per-profile hashes, residual components, and separate gate decisions.

3.2 Separate propagation-step and grid checks

The reused October runs contain 28 trajectories, distinct from the 28 stored branch profiles. Their core and campaign hashes still match the current propagator. Ten new full-horizon trajectories and a two-trajectory short benchmark were run locally; no scheduler or paid compute was used. All new Dysthe cases use $\epsilon = .025$, $\mu = 2$, $s = 0$, and $0 \leq \xi \leq 20$. The odd perturbation is $F[1 + 10^{-3}(\tau/2)e^{-(\tau/2)^2}]$, projected before propagation.

The step table compares successive runs at fixed $L = 160$, $N = 4096$. Its numbers are maxima over integer samples of ξ, normalized by the fine profile norm.

$\Delta\xi$ pair	Control field	Odd field	Signed shape
0.005 / 0.0025	5.96×10^{-6}	5.96×10^{-6}	2.08×10^{-10}
0.0025 / 0.00125	2.96×10^{-7}	2.96×10^{-7}	7.90×10^{-12}
0.00125 / 0.000625	1.66×10^{-8}	1.66×10^{-8}	4.65×10^{-13}

The final-control observed orders are 4.335 and 4.154. At the smallest historical pair, the

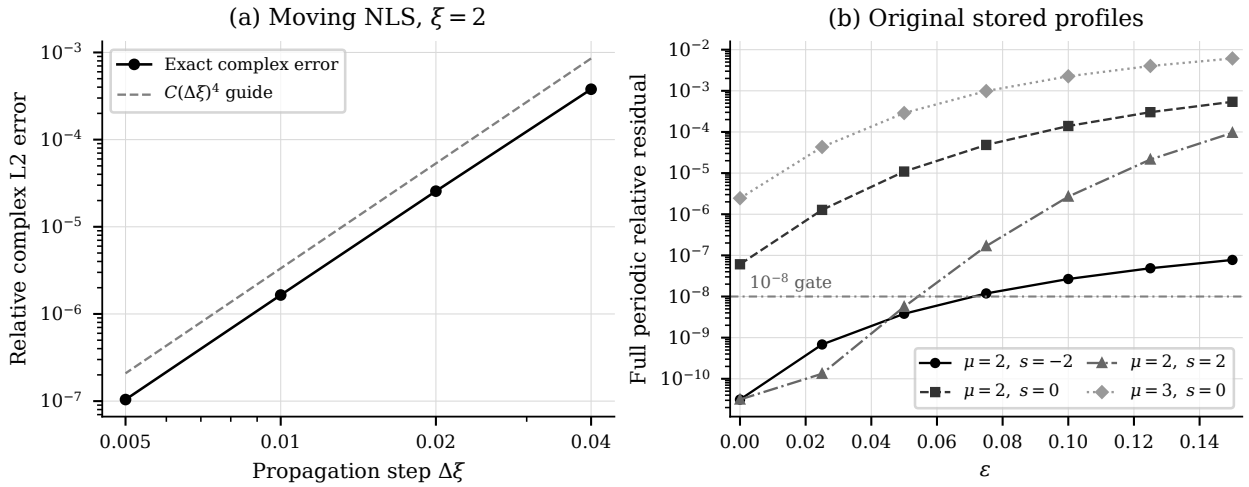


Figure 1: Analytic propagation control and original-profile full-equation audit. The dashed fourth-order guide is scaled for visibility. The profile gate uses the full residual of (2.3), not the iteration residual.

maximum raw-field difference is 1.66×10^{-8}. Conservation cannot expose this phase-sensitive error by itself.

Fixed-domain grid pair	Odd field	Signed shape
$L = 160$: 4096/8192	2.45×10^{-14}	1.51×10^{-14}
$L = 320$: 8192/16384	2.67×10^{-13}	2.31×10^{-14}

Both grid comparisons fix $\Delta\xi = .000625$; Fourier interpolation puts each pair on the finer grid. Thus enlarging the domain is not being confused with raising the retained physical wavenumber cutoff.

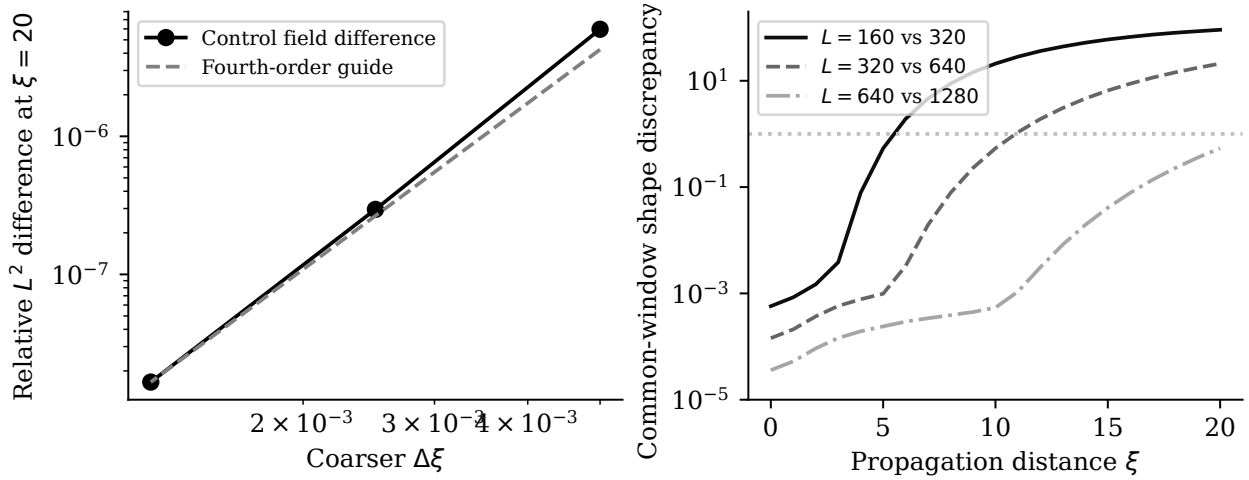


Figure 2: Step convergence and signed weak-response domain discrepancies. The domain panel uses the fine response norm on the common window; the dotted line is a one-percent comparison target, not a rigorous bound.

3.3 Domain sensitivity and the weak odd response

The domain sequence doubles $L = 160, 320, 640, 1280$ at fixed $L/N = 160/4096$ and $\Delta\xi = .000625$. Each periodic profile is recomputed. The table gives results at $\xi = 20$; shape and core differences are normalized by the fine carrier profile on the common window, and the last column divides by the weak shape on that window. The core is $|\tau| < 20$.

L pair	Initial profile	Signed shape	Core	Shape ratio
160 / 320	1.54×10^{-6}	2.19×10^{-5}	8.76×10^{-6}	91%
320 / 640	3.85×10^{-7}	6.69×10^{-6}	2.60×10^{-7}	21%
640 / 1280	9.63×10^{-8}	1.75×10^{-7}	1.64×10^{-10}	0.538%

The previously quoted 67.4% discrepancy for 160/320 used the entire fine-domain response as denominator. The common-window denominator in this table gives 91.1%. These are the same field discrepancy with different denominators, not conflicting results. For 320/640 the full-domain denominator gives 20.6%, compared with 21.0% on the common window.

The 320/640 response outside the smaller period has normalized norm 6.69×10^{-6}, nearly equal to the discrepancy. Periodizing that exterior reduces the discrepancy to 1.95×10^{-7}. Together with the response spectrum and travel times, this strongly supports radiation wrapping as the principal mechanism. Periodic nonlocal profile differences and later nonlinear interactions remain present; folding is only a diagnostic.

At $L = 1280$, the final raw perturbation/control distance is 2.36×10^{-4}, while the shape distance after fitting is 3.25×10^{-5}. The fitted translation is 2.74×10^{-4} and constant phase 6.77×10^{-5} radians. Translation and phase therefore account for most of the raw response, while the remaining signed deformation still includes radiation.

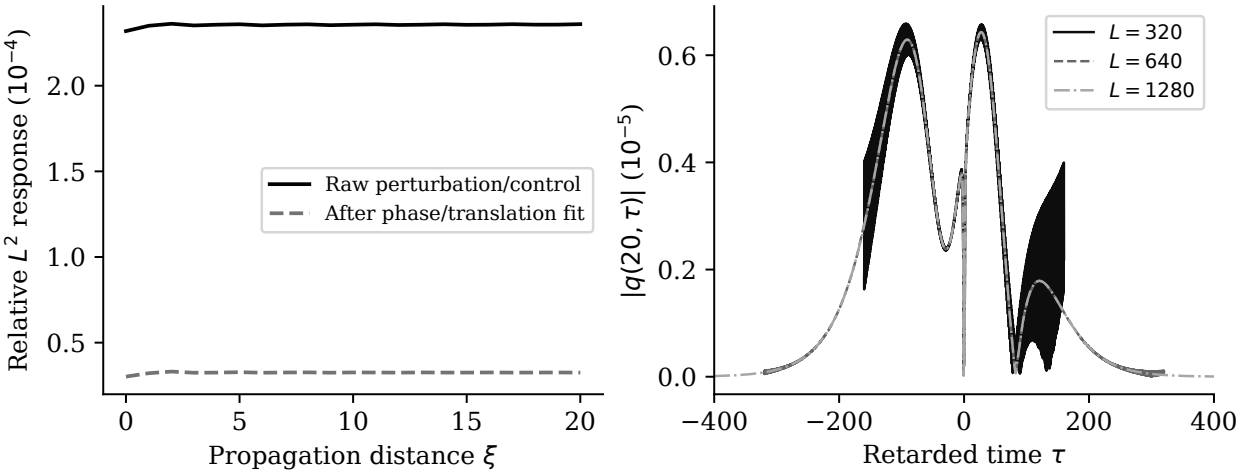


Figure 3: Raw versus fitted response and the final signed-response magnitude. Nearly equal scalar shape norms can hide substantially different radiation fields; the convergence tests compare the complex fields themselves.

3.4 Residuals, invariants, and resources

The largest initial/final discarded-RHS ratio in the accepted-spacing, small-step cases is 9.29×10^{-12}. This is an instantaneous resolution diagnostic, not a time-integrated error bound. The following

invariant maxima cover both control and odd trajectories; energy and momentum are the continuum functionals rather than assumed exact discrete invariants.

L	Action drift	Energy drift	Momentum drift
160	8.66×10^{-13}	2.72×10^{-12}	6.74×10^{-11}
640	6.29×10^{-13}	1.98×10^{-12}	6.74×10^{-11}
1280	7.30×10^{-13}	2.30×10^{-12}	6.73×10^{-11}

The five new full-horizon cases used 809.2 process-wall seconds in total, with peak recorded process RSS 177.9 MiB. Some cases ran concurrently, so these durations are not elapsed session time. The benchmark used less than one second. Every saved field was finite. The existing large-window sech initial guess emitted one benign cosh-overflow warning in its vanishing tails; it did not produce nonfinite profiles or trajectories.

3.5 Recommendation and accuracy limits

Go for the next small initial-data simulator, after review, using this periodic equation and per-case qualification. The reference passes analytic controls, independent residual checks, fourth-order step convergence, and independent grid convergence. No change to the equation or propagator was needed. The added diagnostics separate symmetry drift and identify the domain failure mechanism.

For an illustrative one-percent signed weak-response agreement target, the sampled horizons of the smaller windows are $\xi \leq 5$ for $L = 160$, $\xi \leq 10$ for $L = 320$, and $\xi \leq 20$ for $L = 640$, each assessed against the next doubled domain. These are measured sample-wise agreements for this perturbation, not rigorous error bars or universal domain choices.

The final 640/1280 common-window shape discrepancy is 1.75×10^{-7}, or 0.538% of this weak shape response. A full-domain initial-data simulator must preserve phase, translation, and radiation and expose numerical-resolution checks. New focusing, modulation, and bandwidth ranges need their own convergence evidence.

No-go for a large training corpus or FNO training now. The original $L = 160$, $\xi = 20$ setup is not qualified as an isolated weak-radiation reference. The entire legacy profile library is not qualified either. The next stage should implement a small plug-and-play simulator with a finite, explicitly tested input range; dataset generation and learning require a later decision. Short persistence does not establish soliton discovery or stability.

4 Reproducibility

The checked snapshot is `stage1/source/`; additive modules are installed in the original project's `dysthe-core` and `dysthe-experiments` repositories. The installation manifest lists all nine added files and hashes. Initial provenance, effective configurations, SHA-256 hashes, environment versions, timings, and sampled complex fields accompany the results. Existing unrelated changes and historical arrays are preserved.

Full commands are recorded in `stage1/README.md`. From the task root, use the two snapshot source directories and the existing project's `dysthe-learning/src` on `PYTHONPATH`. Run `pytest -import-mode=importlib -p no:cacheprovider`, followed by these experiment modules (`project` denotes the verified project):

```
python3 -m dysthe_experiments.reference_controls \
  --profiles "$project/results/nonlinear" \
  --output stage1/results/analytic-checks.json
python3 -m dysthe_experiments.reference_stage_one \
  --profile "$profile" --output stage1/results/benchmark --benchmark
python3 -m dysthe_experiments.reference_stage_one \
  --profile "$profile" --output stage1/results/campaign
python3 -m dysthe_experiments.reference_stage_one \
  --profile "$profile" --extended-domain --output stage1/results/extended
python3 -m dysthe_experiments.reference_analysis \
  --historical "$actualruns" --campaign stage1/results/campaign \
  --extended stage1/results/extended --output stage1/results/analysis
python3 stage1/make_report_results.py
```

The initial-data simulator needs new per-family qualification, especially for focusing and radiation. This stage contains no training dataset, FNO training, paid compute, or HPC submission. Its finite periodic evidence also supplies no Euler-equation or experimental validation.

References

[1] F. Fedele and D. Dutykh, *Hamiltonian description and traveling waves of the spatial Dysthe equations*, arXiv:1110.3605v3, 12 April 2012. `https://arxiv.org/html/1110.3605v3`.